Session 1 — Reference Handout

Azure Core Architecture & Services

BFL Azure & AI Foundry Training Program · Part 1 — Azure Cloud
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1. Azure Global Infrastructure

1.1 Geography

A **Geography** is a discrete market — typically a country or defined area — that contains one or more Azure Regions.

- Preserves **data residency, sovereignty, and compliance boundaries**
- Customer data stays within the chosen geography unless explicitly configured otherwise
- Examples: United States, Europe, India, Australia, Canada, Japan, Brazil
- **This is usually the first design decision**, made before region or service selection

1.2 Region

A **Region** is a set of datacenters within a latency-defined perimeter, connected by a dedicated low-latency network.

- Each region has a unique name/code (e.g. `East US`, `Central US`, `Central India`)
- **Not all services are available in every region** — always verify before designing
- Selection criteria: latency to users, compliance/residency, service availability, cost, availability of a paired region

1.3 Region Pair

Most regions are paired with another region **≥ 300 miles away**, within the **same geography**.

Protection	What it means
Physical isolation	Reduces the chance one disaster affects both regions
Sequential updates	Microsoft never updates both paired regions at the same time
Priority recovery	One region in the pair is prioritized for restoration during a broad outage
Residency preserved	Pairing stays inside the same geography

Example pairs: East US ↔ West US · Central India ↔ South India

1.4 Availability Zone (AZ)

Physically separate locations within an AZ-enabled region, each with **independent power, cooling, and networking**.

- Minimum of **three** separate zones per AZ-enabled region
- Connected via high-speed private fiber (low enough latency for synchronous replication)
- Protects against a **single-datacenter failure** — not a region-wide event
- Not every region is AZ-enabled — verify before designing a zone-redundant architecture

1.5 The Nesting Model

```
Geography
   └── Region
          └── Availability Zones (3+)
                 └── Datacenters
```

Region Pairs sit *alongside* this hierarchy, connecting two Regions within the same Geography for disaster recovery.

Rule of thumb: Region Pair = protection against a wide-area disaster. Availability Zone = protection against a single-datacenter failure. Robust designs often use both.

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2. Azure Resource Manager (ARM) & Resource Organization

2.1 What ARM Is

Azure Resource Manager is the **single deployment and management API layer**. Every tool — Portal, CLI, PowerShell, SDKs, Terraform, Bicep — goes through the same ARM API.

- Supports **declarative deployment** (ARM JSON templates or Bicep)
- Applies RBAC, Tags, Azure Policy, and Resource Locks **consistently**
- Every request is authenticated and authorized before ARM acts
- Understands resource dependencies for idempotent, repeatable deployments

2.2 The Scope Hierarchy

```
Management Group   → governs many Subscriptions
   └── Subscription     → billing & access boundary
          └── Resource Group  → logical container, shared lifecycle
                 └── Resource      → the actual building block (VM, storage, DB,
VNet...)
```

Governance flows top-down. Deployment happens bottom-up.
Policy, RBAC, and Tags assigned at a higher scope are **inherited automatically** by everything beneath it.

2.3 Management Groups

- Manage access, policy, and compliance across many subscriptions at once
- Multi-level hierarchy beneath a single **root management group**
- Ideal for org-wide guardrails: allowed regions, allowed resource types, mandatory tagging

- Typical pattern: separate groups for Production / Non-Production / Sandbox

2.4 Subscriptions

- The **billing boundary** and the **access-management boundary**
- Linked to one Microsoft Entra ID tenant for identity
- Default service quotas/limits apply per subscription
- Common pattern: separate subscriptions per environment (Dev/Test/Prod) or business unit

2.5 Resource Groups & Resources

- A **Resource Group** holds the related resources for one solution
- A resource belongs to **exactly one** Resource Group at a time
- Resources share the RG's **lifecycle** — deleting the RG deletes everything inside it
- RBAC, Policy, Tags, and Locks can be applied at the RG level
- The RG has a region for its own metadata, but the resources inside can span multiple regions

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3. Azure Portal — Quick Reference

Feature	What it does
Dashboards & Favorites	Personalized views per user
Resource blades	Consistent create/manage/monitor experience per resource type
Global search	Jump to any resource, service, or docs page
Cloud Shell	Browser-based, pre-authenticated Bash/PowerShell
Azure Resource Graph Explorer	Query resources at scale (KQL-style syntax)
Cost Management + Billing	Built-in cost tracking and budgets
Azure Advisor	Built-in recommendations

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4. Compute Services — Decision Table

Service	Control Level	Best For
Virtual Machines	Full OS control (IaaS)	Lift-and-shift, custom OS or licensed software
App Service	Managed web hosting (PaaS)	Web apps & APIs — no infrastructure to patch
Azure Functions	Serverless, event-driven	Short-lived code triggered by an event
Azure Kubernetes Service (AKS)	Managed Kubernetes	Container orchestration & microservices at scale
Container Instances	Serverless containers	Simple, single containers & burst workloads

Quick trigger phrase: "*event-triggered + no servers to manage*" → think **Azure Functions** first.

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5. Storage Services — Decision Table

Service	Data Shape	Typical Use
Blob Storage	Unstructured objects	Images, video, backups, data-lake content
Azure Files	File share (SMB/NFS)	Mountable network drive
Queue Storage	Messages	Decoupling application components
Table Storage	NoSQL key-value	Fast semi-structured lookups
Managed Disks	Block storage	VM OS and data disks

Redundancy Options (increasing resilience)

Option	Replication scope
LRS	Within a single datacenter
ZRS	Synchronously across Availability Zones in one region
GRS	Asynchronously to a paired region
GZRS	Zone-redundant locally + geo-replicated to a paired region

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6. Core Architecture — The Pattern That Repeats

```
EDGE & ACCESS           Users, DNS, Front Door/Gateway, Identity (Entra ID)
APPLICATION SERVICES    App Service, Functions, AKS, Container Instances
DATA & STORAGE          Blob, Files, Managed Disks, Databases
FOUNDATION SERVICES     Key Vault, Monitor, Policy, Backup, RBAC
```

Every layer sits inside a **Resource Group**, inside a **Subscription**, inside a **Region** chosen for compliance and latency. This same four-layer pattern reappears with an AI layer added on top in Part 2 — AI Foundry.

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7. Key Takeaways

1. **Geography** → **Region** → **Availability Zone** drives compliance and resilience decisions.
2. **Region Pairs** protect against wide-area disasters; **Availability Zones** protect against single-datacenter failure.
3. **ARM** is the one API behind every Azure tool.
4. **Management Group** → **Subscription** → **Resource Group** → **Resource** — govern top-down, deploy bottom-up.
5. **Portal, Compute, and Storage** are the day-to-day building blocks of every Azure architecture.

References (Microsoft Learn)

- Regions, region pairs & Availability Zones — `learn.microsoft.com/azure/reliability/availability-zones-overview`
- Azure Resource Manager overview — `learn.microsoft.com/azure/azure-resource-manager/management/overview`
- Organize resources with Management Groups — `learn.microsoft.com/azure/governance/management-groups/overview`
- Azure Portal overview — `learn.microsoft.com/azure/azure-portal/azure-portal-overview`
- Compare Azure compute options — `learn.microsoft.com/azure/architecture/guide/technology-choices/compute-decision-tree`
- Introduction to Azure Storage — `learn.microsoft.com/azure/storage/common/storage-introduction`

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